Probability Theory and Statistics
Lecture 10

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„The" Famous Distribution

Consider the following random quantities: shoe size of people, birth weight, blood pressure, height (women/men separately), daily average temperature (on a given day of the year), humidity, measurement error in an experiment, noise.

Many effects are the continuous result of many small, independent factors.

Names: normal distribution, Gaussian distribution, bell curve (its density function).

The most common and most significant distribution in probability theory and statistics.

This part significantly draws on slides by Márta Pintér from three years ago.

Normal Distribution

Definition

A continuous random variable Y has a normal distribution with parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ if its density is

$$f_Y(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad x \in \mathbb{R}.$$

Notation: $X \sim \mathrm{N}(\mu;\sigma^2)$. $\mathrm{N}(0;1)$ is called standard normal.

(Remark: we denote the square root of σ^2 by σ, but the distribution is parameterized by σ^2, not by σ.)

The density is symmetric about μ, and the larger σ is, the more spread out it becomes.

Standard Normal: Density

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{x^2}{2}}, \qquad x \in \mathbb{R}.$$

This is a nonnegative function. The following claim implies that it is indeed a density, i.e., $\int_{-\infty}^{\infty} f_Y(x)\,\mathrm{d}x = 1$:

Theorem

$\int_{-\infty}^{\infty} \mathrm{e}^{-\frac{x^2}{2}}\,\mathrm{d}x = \sqrt{2\pi}.$

(Proof: via polar coordinates)

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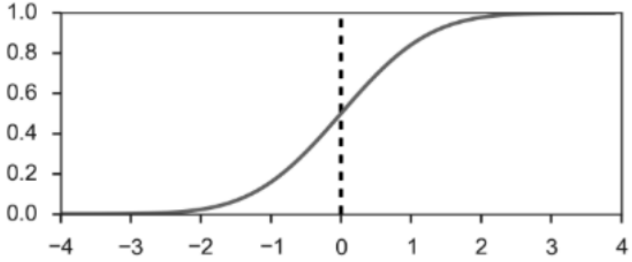
Properties of φ: even function; limit 0 at $\pm\infty$; positive everywhere; maximum at 0.

Standard Normal: Distribution Function

$$\Phi(x) = \int_{-\infty}^{x} \varphi(t)\mathrm{d}t = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{t^2}{2}} \mathrm{d}t.$$

This cannot be expressed in elementary functions.
Instead we use approximations.



Properties of Φ: it has all the properties of a distribution function, it is strictly increasing, and $\Phi(x) = 1 - \Phi(-x)$.

Standard Normal: Table

A standard normális eloszlás eloszlásfüggvényének táblázata

0,01	0,02	0,03	0,04	0,05	0,06	0,07	0,08	0,09
0,5040	0,5080	0,5120	0,5160	0,5199	0,5239	0,5279	0,5319	0,5359
0,5438	0,5478	0,5517	0,5557	0,5596	0,5636	0,5675	0,5714	0,5753
0,5832	0,5871	0,5910	0,5948	0,5987	0,6026	0,6064	0,6103	0,6141
0,6217	0,6255	0,6293	0,6331	0,6368	0,6406	0,6443	0,6480	0,6517
0,6591	0,6628	0,6664	0,6700	0,6736	0,6772	0,6808	0,6844	0,6879
0,6950	0,6985	0,7019	0,7054	0,7088	0,7123	0,7157	0,7190	0,7224
0,7291	0,7324	0,7357	0,7389	0,7422	0,7454	0,7486	0,7517	0,7549
0,7611	0,7642	0,7673	0,7704	0,7734	0,7764	0,7794	0,7823	0,7852
0,7910	0,7939	0,7967	0,7995	0,8023	0,8051	0,8078	0,8106	0,8133
0,8186	0,8212	0,8238	0,8264	0,8289	0,8315	0,8340	0,8365	0,8389
0,8438	0,8461	0,8485	0,8508	0,8531	0,8554	0,8577	0,8599	0,8621
0,8665	0,8686	0,8708	0,8729	0,8749	0,8770	0,8790	0,8810	0,8830
0,8869	0,8888	0,8907	0,8925	0,8944	0,8962	0,8980	0,8997	0,9015
0,9049	0,9066	0,9082	0,9099	0,9115	0,9131	0,9147	0,9162	0,9177
0,9207	0,9222	0,9236	0,9251	0,9265	0,9279	0,9292	0,9306	0,9319
0,9345	0,9357	0,9370	0,9382	0,9394	0,9406	0,9418	0,9429	0,9441
0,9463	0,9474	0,9484	0,9495	0,9505	0,9515	0,9525	0,9535	0,9545
0,9564	0,9573	0,9582	0,9591	0,9599	0,9608	0,9616	0,9625	0,9633
0,9649	0,9656	0,9664	0,9671	0,9678	0,9686	0,9693	0,9699	0,9706
0,9719	0,9726	0,9732	0,9738	0,9744	0,9750	0,9756	0,9761	0,9767
0,9778	0,9783	0,9788	0,9793	0,9798	0,9803	0,9808	0,9812	0,9817
0,9826	0,9830	0,9834	0,9838	0,9842	0,9846	0,9850	0,9854	0,9857
0,9864	0,9868	0,9871	0,9875	0,9878	0,9881	0,9884	0,9887	0,9890
0,9896	0,9898	0,9901	0,9904	0,9906	0,9909	0,9911	0,9913	0,9916
0,9920	0,9922	0,9925	0,9927	0,9929	0,9931	0,9932	0,9934	0,9936
0,9940	0,9941	0,9943	0,9945	0,9946	0,9948	0,9949	0,9951	0,9952
0,9955	0,9956	0,9957	0,9959	0,9960	0,9961	0,9962	0,9963	0,9964
0,9966	0,9967	0,9968	0,9969	0,9970	0,9971	0,9972	0,9973	0,9974
0,9975	0,9976	0,9977	0,9977	0,9978	0,9979	0,9979	0,9980	0,9981
0,9982	0,9982	0,9983	0,9984	0,9984	0,9985	0,9985	0,9986	0,9986
0,9987	0,9987	0,9988	0,9988	0,9989	0,9989	0,9989	0,9990	0,9990
0,9991	0,9991	0,9991	0,9992	0,9992	0,9992	0,9992	0,9993	0,9993
0,9993	0,9994	0,9994	0,9994	0,9994	0,9994	0,9995	0,9995	0,9995
0,9995	0,9995	0,9996	0,9996	0,9996	0,9996	0,9996	0,9996	0,9997
0,9997	0,9997	0,9997	0,9997	0,9997	0,9997	0,9997	0,9997	0,9998

x	0,00	0,01	0,02
0,0	0,5000	0,5040	0,5080
0,1	0,5398	0,5438	0,5478
0,2	0,5793	0,5832	0,5871
0,3	0,6179	0,6217	0,6255
0,4	0,6554	0,6591	0,6628
0,5	0,6915	0,6950	0,6985

$$\Phi(0{,}21) = 0{,}5832$$

$$\Phi(0{,}315) = 0{,}6236$$

$$\Phi(-0{,}11) = 1 - 0{,}5438 = 0{,}4562$$

Standard Normal: Table

„In practice sessions in earlier years, the following procedure emerged for using the table:
a) If we know x to more than two decimal places and $\Phi(x)$ is asked for, then round to the nearest value with two decimals; compute $\Phi(x)$ for that, and if x is exactly halfway, then average the Φ-values of the two neighboring two-decimal x's.
b) If you need to look up a value in reverse from the normal table, i.e., you know y with $\Phi(x) = y$, but y lies between two table entries, i.e., $\Phi(x_1) < y < \Phi(x_2)$ where x_1, x_2 are adjacent in the table, then take the closer x as the inverse value; if y is exactly halfway, then take the smaller x."

Non-Standard Normal

The functions given for the non-standard normal distribution are also densities.
Indeed, a non-standard normal is obtained from the standard by a linear transformation:

Let $\mu, \sigma \in \mathbb{R}$, $\sigma > 0$, let X be a continuous random variable, whose distribution function is differentiable at every point, and let $Y = \sigma X + \mu$. Then

$$f_Y(x) = \frac{1}{\sigma} f_X\left(\frac{x-\mu}{\sigma}\right).$$

(Proof)

If $X \sim \mathrm{N}(0;1)$, then $Y = \sigma X + \mu$ is $\sim \mathrm{N}(\mu; \sigma^2)$, because

$$\frac{1}{\sigma} f_X\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \mathrm{e}^{-\frac{(\frac{x-\mu}{\sigma})^2}{2}} = \frac{1}{\sqrt{2\pi\sigma^2}} \mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad x \in \mathbb{R}.$$

Standardization

Conversely: let $Y \sim \mathrm{N}(\mu; \sigma^2)$ and let $X = \frac{Y-\mu}{\sigma}$. This has the standard normal distribution, since by the previous calculation

$$f_X(x) = \sigma f_X(\sigma x + \mu) = \sigma \frac{1}{\sqrt{2\pi\sigma^2}} \mathrm{e}^{-\frac{(\sigma x+\mu-\mu)^2}{2\sigma^2}} = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{x^2}{2}}.$$

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Thus we have shown:

Theorem

Let $\mu \in \mathbb{R}$ and $\sigma > 0$. A random variable Y has distribution $\mathrm{N}(\mu; \sigma^2)$ if and only if there exists $X \sim \mathrm{N}(0; 1)$ such that $Y = \sigma X + \mu$, i.e., $X = \frac{Y-\mu}{\sigma}$.

Expectation and Variance of the Normal

Theorem

Let $Y \sim \mathrm{N}(\mu; \sigma^2)$. Then $\mathbb{E}(Y) = \mu$ and $\mathbb{D}^2(Y) = \sigma^2$.

For the proof we first show it suffices to verify the claim for the standard normal ($\mu = 0$, $\sigma = 1$).
Indeed, assume that case has been proved, and let $Y \sim \mathrm{N}(\mu; \sigma^2)$. Then there exists $X \sim \mathrm{N}(0; 1)$ with $Y = \sigma X + \mu$, hence

$$\mathbb{E}(Y) = \mathbb{E}(\sigma X + \mu) = \sigma \mathbb{E}(X) + \mu = \mu$$

and

$$\mathbb{D}^2(Y) = \mathbb{D}^2(\sigma X + \mu) = \sigma^2 \mathbb{D}^2(X) = \sigma^2.$$

(Proof for $X \sim \mathrm{N}(0; 1)$: $\mathbb{E}(X) = 0$, $\mathbb{E}(X^2) = 1$ (by integration by parts), see lecture/notes.)

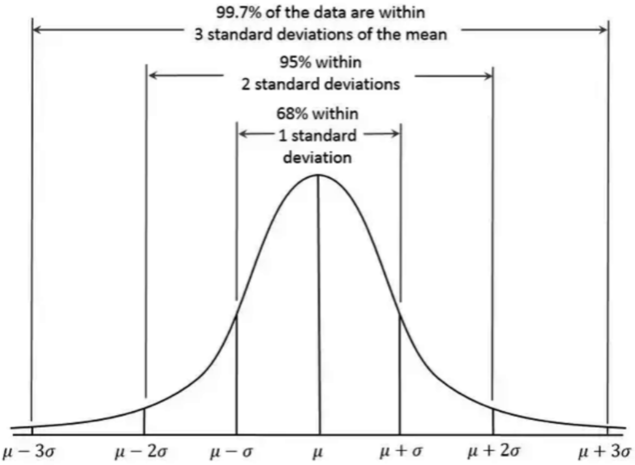
Standardization

Standardizing the normal: if $Y \sim \mathrm{N}(\mu; \sigma^2)$, then

$$\frac{Y - \mathbb{E}(Y)}{\mathbb{D}(Y)} \sim \mathrm{N}(0; 1).$$

Remark: We can also standardize other distributions (by subtracting the expectation and dividing by the standard deviation), provided the second moment is finite. This standardized variable will typically not be standard normal, but it will have expectation 0 and standard deviation 1.

Variance of the Normal, Visually



- About 68% of the probability mass lies in the interval $[\mu - \sigma, \mu + \sigma]$,
- f_Y changes convexity at the points $\mu \pm \sigma$.

Normal Distribution

Example: We try to transmit a ± 1-valued signal through a noisy channel. Because of the noise, the received value is not necessarily ± 1, but the true signal value plus a $\mathrm{N}(0;\sigma^2)$ random variable. Determine σ if the probability that a „$+1$" is received as a negative value is 0.0226.

Noise: $Y \sim \mathrm{N}(0;\sigma^2)$.

$$\mathbb{P}(1+Y<0)=\mathbb{P}(Y<-1)=\mathbb{P}\Big(\frac{Y}{\sigma}<\frac{-1}{\sigma}\Big)=\Phi\Big(\frac{-1}{\sigma}\Big)=0.0226,$$

thus $\sigma=\frac{1}{2.00}$.

Normal Distribution

Example: The sample temperature (°C) is $Y \sim \mathrm{N}(-2; 1.69)$.
What is the probability that the sample temperature exceeds $0°$C?

Standardize: $X = \frac{Y+2}{\sqrt{1.69}} \sim \mathrm{N}(0; 1)$.

$$\mathbb{P}(Y > 0) = \mathbb{P}\left(\frac{Y+2}{1.3} > \frac{2}{1.3}\right) = 1 - \Phi\left(\frac{2}{1.3}\right).$$

$\mathbb{P}(Y > 0) \approx 0.0620 \approx 6\%$

Aside: Convolution

If X and Y are independent random variables, then the distribution of $X+Y$ is called the convolution of X and Y. In the continuous case, the density of $X+Y$ is

$$f_{X+Y}(x) = \int_{-\infty}^{\infty} f_X(z)\, f_Y(x-z)\, \mathrm{d}z, \qquad x \in \mathbb{R}.$$

- The binomial distribution is the sum of independent indicators.
- If $X \sim \mathrm{Pois}(\lambda)$ and $Y \sim \mathrm{Pois}(\mu)$, then $X+Y \sim \mathrm{Pois}(\lambda+\mu)$.
- Sums of $\mathrm{Geo}(p)$ variables give the "negative binomial" distribution.
- Sum of uniform distributions: the Irwin–Hall distribution (see figure).
- Sum of exponentials: the "gamma" distribution.
- If $X \sim \mathrm{N}(\mu_1; \sigma_1^2)$ and $Y \sim \mathrm{N}(\mu_2; \sigma_2^2)$, then $X+Y \sim \mathrm{N}(\mu_1+\mu_2; \sigma_1^2+\sigma_2^2)$.

de Moivre–Laplace Theorem

Question: Why does the normal distribution appear in applications and measurements?
The following theorem says that the standardized binomial can be approximated by the normal when p is fixed and n is large:

Theorem (de Moivre–Laplace)

Let $p \in (0,1)$ and $S_n \sim \mathrm{B}(n;p)$. Then for all real $a < b$,

$$\lim_{n\to\infty} \mathbb{P}\left(a < \frac{S_n - \mathbb{E}(S_n)}{\mathbb{D}(S_n)} < b\right) = \int_a^b \varphi(x)\mathrm{d}x = \Phi(b) - \Phi(a).$$

As we have seen: $\dfrac{S_n - \mathbb{E}(S_n)}{\mathbb{D}(S_n)} = \dfrac{S_n - np}{\sqrt{np(1-p)}}$.

Remark: One can say something concrete about the speed of convergence as well.

de Moivre–Laplace Theorem

Theorem

Let $p \in (0,1)$ and $S_n \sim \mathrm{B}(n;p)$. Then for all real $a < b$,

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For the Poisson distribution we saw that the binomial limit distribution is Poisson.
Now: the binomial limit distribution is normal.

- There we did not subtract an (n-dependent) mean; here we do.
- There we did not divide by an (n-dependent) standard deviation; here we do.
- There we assumed $p_n \to 0$ and $np_n \to \lambda$ (e.g., $p_n = \lambda/n$), while here p is constant.
  Thus: n is large in both approximations, but for Poisson we have $p = p_n \to 0$, whereas for the normal approximation/CLT we do not.

Example: de Moivre–Laplace

Of all mathematicians, 31.4% wear sandals. Out of one hundred randomly chosen mathematicians, what is the approximate probability that we find fewer than 25 pairs of sandals on them?

Number of sandal pairs: $S_n \sim \mathrm{B}(n;p)$ with $n = 100$, $p = 0.314$.

$\mathbb{E}(S_n) = 31.4, \quad \mathbb{D}(S_n) = \sqrt{np(1-p)} = \sqrt{100 \cdot 0.314 \cdot 0.686} \approx 4.6412.$

$\dfrac{S_n - 31.4}{4.6412}$ is approximately $\mathrm{N}(0;1)$.

$\mathbb{P}(S_n < 25) = \mathbb{P}\Big(\dfrac{S_n - 31.4}{4.6412} < \dfrac{25 - 31.4}{4.6412}\Big) \approx \Phi(-1.3790) =$
$1 - \Phi(1.3790) \approx 0.0839 \approx 8\%.$

Poisson vs. Normal

Not only the binomial, but e.g. the standardized Poisson distribution also approximates the normal distribution if we repeat the Poisson experiment many times.